



IndianOil

JOURNEY RISK MANAGEMENT (JRM) STUDY

Salem Terminal TO SRI VENKATESHWARA FUELS

Objective of the JRM Report

This JRM report is designed to ensure compliance with the Central Motor Vehicle Rules, 1989 (CMVR), AIS 140 standards, and the Road Transport Safety Policy (RTSP). It provides a comprehensive risk assessment for the transportation of hazardous materials along specified routes. By integrating these legal frameworks, the report offers a broad strategy for identifying and mitigating route-specific risks.

Regulatory Compliance

The report complies with the Central Motor Vehicles (Eleventh Amendment) Rules, 2022, mandating safe transportation practices for N2 and N3 category vehicles carrying hazardous materials. These rules require detailed route assessments, especially regarding road conditions, speed limits, and risk areas, to ensure safety compliance.

Risk Management Strategy

This report categorizes transportation routes into high-risk and medium-risk areas, with a focus on factors such as sharp turns, accident-prone regions, and elevation changes. The goal is to provide actionable

recommendations to minimize these risks, including speed regulations, driver warnings for hazardous zones, and the option of alternate routes.

Compliance with the Road Transport Safety Policy (RTSP)

The report integrates RTSP provisions, including mandatory driving hours, rest periods, and nighttime driving restrictions. It ensures that drivers follow official guidelines, such as taking prescribed rest breaks and avoiding dangerous road conditions like poor visibility, heavy crowds, or high-traffic areas during peak hours.

Emergency Preparedness and Response

The report highlights the significance of predetermined emergency stops for refueling, rest, and overnight stays. It includes protocols for safe responses to road hazards, alternative routes, and rerouting processes if roads are closed or severe weather arises. This aligns with the RTSP emphasis on driver safety and rapid emergency response.

Environmental Considerations

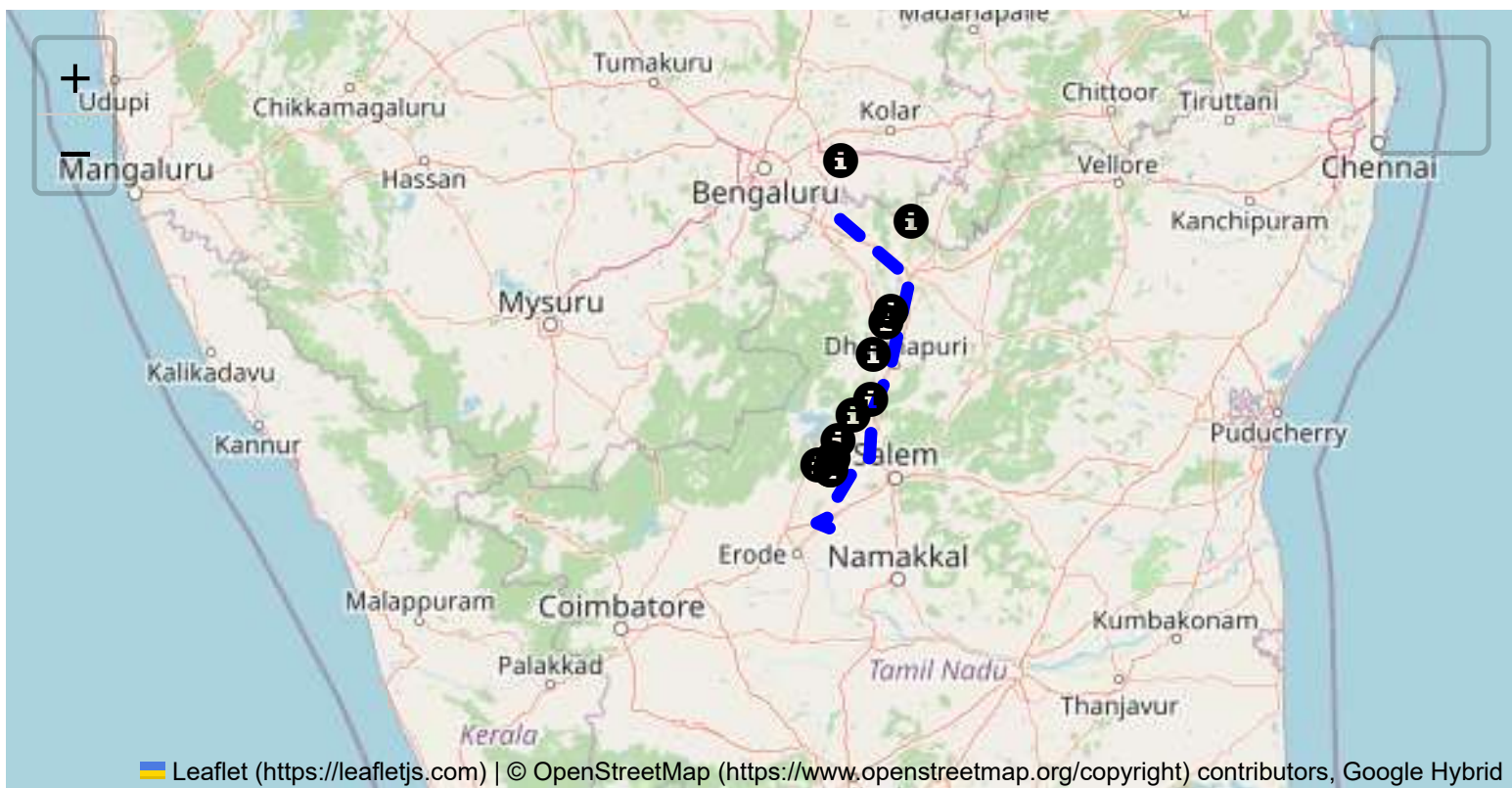
The JRM report addresses environmental risks along the route, ensuring compliance with environmental protection laws in ecologically sensitive zones. It suggests strategies such as identifying areas near water bodies, forests, or populated regions and implementing safety measures to minimize environmental impacts during transport.

Journey Risk Mitigation

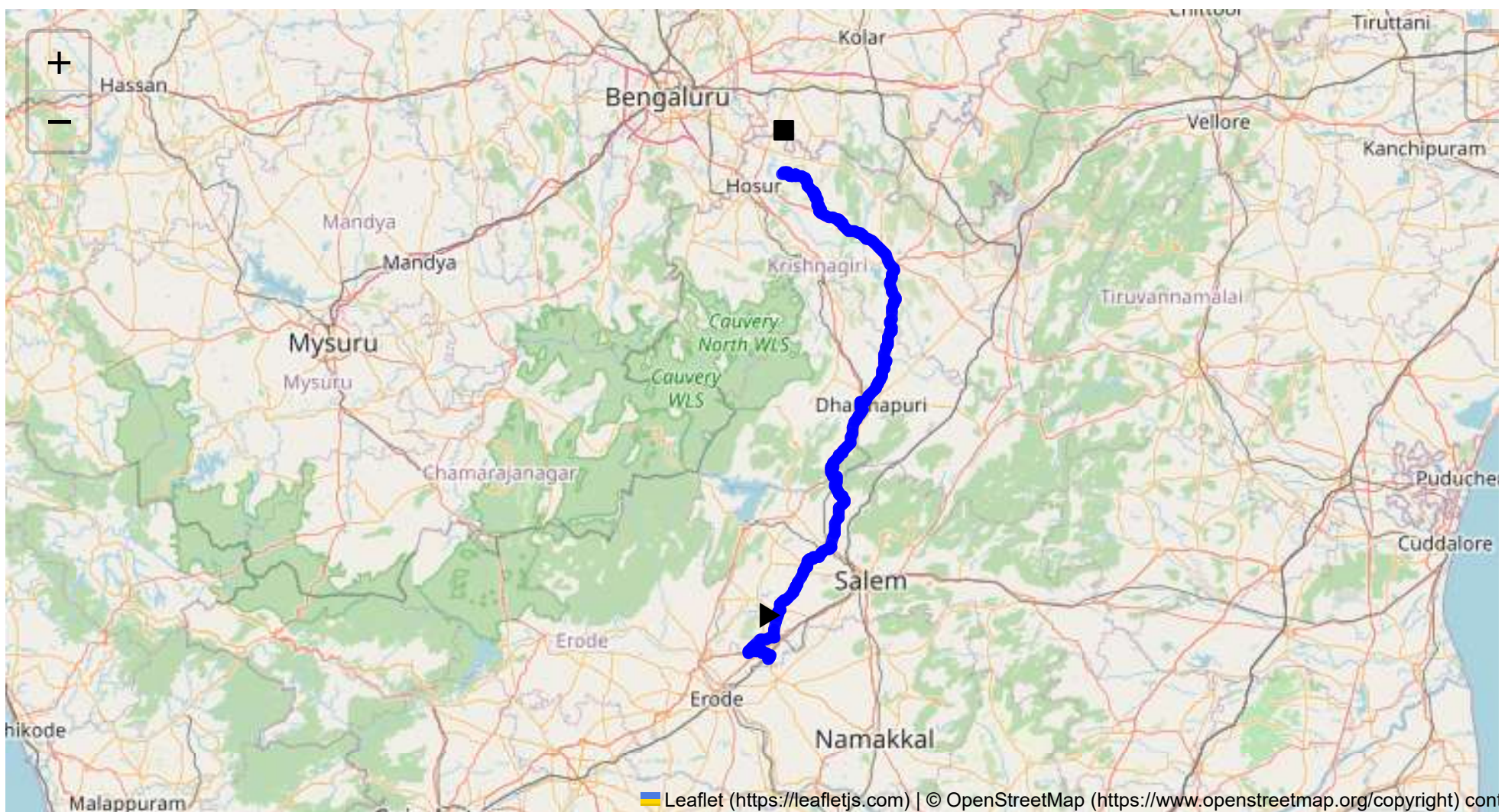
The report includes route-specific risk assessments, detailed journey charts, and defensive driving guidelines for each transport route. Integration with vehicle tracking systems guarantees real-time warnings on hazardous areas, speed limits, and mandatory stops, consistent with RTSP and CMVR safety norms.

Compliance with Government Directives

This report fully adheres to governmental directives regarding hazardous material transportation, implementing mandatory speed limits, nighttime driving restrictions, and comprehensive driver briefings and real-time alerts about route-related risks.



Route Summary:
Total Distance: 200.78 km
Estimated Duration: 4.1 hours
Adjusted Duration (Heavy Vehicle): 5.1 hours
Start: (11.4381, 77.8734)
End: (12.762679, 77.911034)



Welcome to the Journey Risk Management Study

1. Overview of the Route Map

This route starts in Sangagiri and ends in Avalapalli, Hosur, spanning approximately 200.78 kilometers, passing through multiple towns and cities in Tamil Nadu. Key waypoints include Puthur, Padaiveedu, Sangagiri, Erumaipatti, Desavilakku, Omalur, Thoppur, Dharmapuri, Virupakshipuram, and Krishnagiri.

The journey involves a mix of state highways (e.g., SH 86) and main roads, providing a relatively direct path to the final destination.

2. Typical Weather Conditions and Potential Weather-Related Hazards

Tamil Nadu generally experiences hot and dry weather, with temperatures potentially reaching very high levels during the summer months (April to June). The monsoon season (June to September) can bring heavy rains, leading to reduced visibility and potential flooding in low-lying areas. During the Northeast Monsoon (October to December), expect occasional heavy rains, which might affect road conditions.

3. Analysis of Traffic Patterns

Traffic tends to be heavier in the mornings (7:00-10:00 AM) and late afternoons to evenings (4:00-8:00 PM), particularly near urban centers like Salem and Krishnagiri. Congestion-prone areas include main intersections and market areas in these cities. Local agricultural activity may also contribute to traffic, especially during harvest seasons.

4. Assessment of Road Quality and Infrastructure

The road quality varies along the route. Major highways like Salem-Kochi Highway and parts of NH44 are generally well-maintained, with multiple lanes and clear signage. However, some rural roads and smaller highways may have potholes, uneven surfaces, or be narrow, especially during the monsoon season when maintenance can be delayed.

5. Suggestions for Alternative Routes for Emergencies

In emergencies, consider using the Bangalore-Chennai Highway (NH 48) for bypassing congested or flood-prone areas. Local detours through village roads might be necessary, but verify road conditions as rural routes can be poorly maintained.

6. Summary of Local Regulations Affecting Hazardous Material Transport

Transport of hazardous materials is regulated by strict guidelines. Drivers need a valid license for dangerous goods, and vehicles must adhere to safety specifications. Specific zones, especially residential and school areas, may have restrictions on transporting hazardous materials during certain hours.

7. Overview of Historical Incidents

There have been occasional incidents of heavy vehicle accidents in dense traffic areas, most notably during inclement weather conditions. Reports of hazardous materials spillage are rare but highlight the need for caution when navigating narrow or congested roads.

8. Environmental Considerations and Sensitive Areas

The route traverses agricultural regions that are sensitive to chemical contamination. It is crucial to prevent leaks and spills, especially in ecologically sensitive zones near water bodies or habitats.

9. Analysis of Communication Coverage

Most of the route has reliable mobile network coverage, but rural stretches, such as forested areas near Thoppur, may have dead zones. Using a GPS device with offline maps is advisable.

10. Estimated Emergency Response Times

Emergency response times vary, with more rapid support available in urban areas like Salem and Krishnagiri. Rural response could face delays, taking 30-60 minutes longer due to distance and road conditions.

12. Overall Summary of Risk Assessment

The route from Sangagiri to Hosur through multiple transit points presents moderate risks due to varying road conditions, traffic congestion, and weather impacts. While primary highways are generally safe, vigilance is required in rural and monsoon-affected areas, particularly concerning hazardous materials. Preparation with alternative routes and awareness of regulations and emergency contacts can mitigate these risks.

Risk Assessment - Turns

	Risk Type	Risk Level	Coordinates	Speed Limit
0	Turn	High	11.43961, 77.87341	15 KM/Hr
1	Turn	High	11.44077, 77.87533	15 KM/Hr
2	Turn	High	11.44898, 77.87410	15 KM/Hr
3	Turn	Medium	11.45352, 77.85698	30 KM/Hr
4	Turn	High	11.46303, 77.84945	15 KM/Hr
5	Turn	Medium	11.45545, 77.81812	30 KM/Hr
6	Turn	High	11.45584, 77.81695	15 KM/Hr
7	Turn	Medium	11.45656, 77.81709	30 KM/Hr
8	Turn	High	11.49448, 77.88490	15 KM/Hr
9	Turn	Medium	11.56352, 77.90078	30 KM/Hr
10	Turn	Medium	11.59034, 77.90578	30 KM/Hr
11	Turn	Medium	11.59328, 77.90809	30 KM/Hr
12	Turn	Medium	11.63867, 77.94841	30 KM/Hr

	Risk Type	Risk Level	Coordinates	Speed Limit
13	Turn	High	11.64217, 77.95200	15 KM/Hr
14	Turn	Medium	11.64712, 77.95118	30 KM/Hr
15	Turn	Medium	11.67858, 77.96834	30 KM/Hr
16	Turn	Medium	11.68550, 77.97610	30 KM/Hr
17	Turn	Medium	11.69776, 77.97983	30 KM/Hr
18	Turn	Medium	11.70179, 77.97900	30 KM/Hr
19	Turn	Medium	11.71394, 77.99572	30 KM/Hr
20	Turn	Medium	11.71916, 78.01117	30 KM/Hr
21	Turn	Medium	11.72504, 78.01680	30 KM/Hr
22	Turn	High	11.73766, 78.03445	15 KM/Hr
23	Turn	High	11.73651, 78.03868	15 KM/Hr
24	Turn	Medium	11.74120, 78.04588	30 KM/Hr
25	Turn	Medium	11.86805, 78.08353	30 KM/Hr
26	Turn	Medium	11.88084, 78.08108	30 KM/Hr
27	Turn	Medium	11.91159, 78.05857	30 KM/Hr
28	Turn	Medium	11.93307, 78.06426	30 KM/Hr
29	Turn	Medium	11.93545, 78.06130	30 KM/Hr
30	Turn	Medium	11.93694, 78.05425	30 KM/Hr
31	Turn	Medium	11.94867, 78.04680	30 KM/Hr
32	Turn	Medium	11.95648, 78.04768	30 KM/Hr
33	Turn	High	11.95803, 78.04964	15 KM/Hr
34	Turn	Medium	11.96068, 78.04960	30 KM/Hr
35	Turn	High	11.96370, 78.05142	15 KM/Hr
36	Turn	High	11.96650, 78.04927	15 KM/Hr
37	Turn	Medium	12.12094, 78.13269	30 KM/Hr
38	Turn	Medium	12.14025, 78.12819	30 KM/Hr
39	Turn	High	12.14099, 78.12862	15 KM/Hr
40	Turn	High	12.14056, 78.13957	15 KM/Hr
41	Turn	Medium	12.17102, 78.16860	30 KM/Hr
42	Turn	Medium	12.23131, 78.19497	30 KM/Hr
43	Turn	High	12.24611, 78.19519	15 KM/Hr
44	Turn	Medium	12.25200, 78.20034	30 KM/Hr
45	Turn	Medium	12.31895, 78.20727	30 KM/Hr
46	Turn	Medium	12.40026, 78.21775	30 KM/Hr

	Risk Type	Risk Level	Coordinates	Speed Limit
47	Turn	Medium	12.40734, 78.22344	30 KM/Hr
48	Turn	Medium	12.50342, 78.22352	30 KM/Hr
49	Turn	Medium	12.54446, 78.20114	30 KM/Hr
50	Turn	Medium	12.61917, 78.08303	30 KM/Hr
51	Turn	Medium	12.63696, 78.07061	30 KM/Hr
52	Turn	High	12.66145, 78.01543	15 KM/Hr
53	Turn	Medium	12.66919, 78.01683	30 KM/Hr
54	Turn	Medium	12.67465, 78.01432	30 KM/Hr
55	Turn	High	12.67585, 78.01191	15 KM/Hr
56	Turn	Medium	12.70105, 78.00458	30 KM/Hr
57	Turn	Medium	12.70419, 78.00083	30 KM/Hr
58	Turn	Medium	12.71008, 78.00038	30 KM/Hr
59	Turn	Medium	12.71082, 77.99959	30 KM/Hr
60	Turn	Medium	12.71395, 77.99898	30 KM/Hr
61	Turn	Medium	12.71624, 77.99629	30 KM/Hr
62	Turn	Medium	12.71751, 77.99579	30 KM/Hr
63	Turn	High	12.75453, 77.97518	15 KM/Hr
64	Turn	High	12.76098, 77.94755	15 KM/Hr
65	Turn	High	12.75902, 77.94528	15 KM/Hr
66	Turn	Medium	12.75966, 77.93945	30 KM/Hr
67	Turn	High	12.75787, 77.93620	15 KM/Hr
68	Turn	High	12.76227, 77.92663	15 KM/Hr
69	Turn	High	12.76782, 77.92477	15 KM/Hr
70	Turn	Medium	12.76728, 77.92147	30 KM/Hr
71	Turn	Medium	12.76748, 77.91705	30 KM/Hr

Emergency Locations

	type	name	coordinates	speed_limit	risk_level
1	hospital	Sri Shanthi Hospital	11.569272, 77.901937	30 km/h	Medium
2	hospital	Omalur Government Hospital	11.74161, 78.044296	30 km/h	Medium
3	clinic	R.R. Nursing Home	11.7394129, 78.0439861	30 km/h	Medium

	type	name	coordinates	speed_limit	risk_level
4	hospital	Sri Hari Hospital	11.7396726, 78.0440701	30 km/h	Medium
5	hospital	Sugumar Hospital	11.7396031, 78.0426955	30 km/h	Medium
6	hospital	Sri Sugam Hospital	11.7429821, 78.0461039	30 km/h	Medium
7	hospital	Saravana Hospital	11.7409303, 78.0458308	30 km/h	Medium
8	hospital	Arun Hospital	11.7429892, 78.0466881	30 km/h	Medium
9	clinic	Saratha Clinic	11.7422818, 78.0458175	30 km/h	Medium
11	clinic	Sri Meenakshi Nursing Home, Salem	11.7449162, 78.0474637	30 km/h	Medium
12	hospital	Vinayaka Child Hospital	11.745812, 78.0480225	30 km/h	Medium
13	hospital	Dr. G.R.Thilakam Hospital	11.746097, 78.04769	30 km/h	Medium
14	clinic	Divyam Child Care Clinic	11.7447135, 78.0473858	30 km/h	Medium
15	hospital	Thulasi Hospital	11.7461516, 78.0469984	30 km/h	Medium
16	clinic	Momota Clinic	11.744982, 78.047397	30 km/h	Medium
17	hospital	Amirthalingam Hospital	11.7456708, 78.0470548	30 km/h	Medium
18	hospital	Children Hospital	11.7458377, 78.0480205	30 km/h	Medium
19	clinic	Gomathi Clinic	11.7445293, 78.0470401	30 km/h	Medium
20	clinic	Manimuthu Skin Clinic	11.7521189, 78.0484058	30 km/h	Medium
22	hospital	L.R. Hospital	11.8703354, 78.0844554	30 km/h	Medium
23	clinic	JSA Clinic	11.8700619, 78.0844982	30 km/h	Medium
25	hospital	Government Hospital	12.3048002, 78.2060881	30 km/h	Medium
29	hospital	TCR Multi Speciality Hospital	12.5088183, 78.2237657	30 km/h	Medium
30	clinic	Joe Hospital	12.5209601, 78.2150623	30 km/h	Medium
31	clinic	Ohm Prakash Clinic	12.5206321, 78.2159682	30 km/h	Medium
32	clinic	Gokulam Clinic & Scans	12.5207976, 78.2162023	30 km/h	Medium
33	clinic	Srinivasa Clinic	12.520975, 78.215755	30 km/h	Medium
34	clinic	Dr. Anbalagan Clinic	12.5208366, 78.2160593	30 km/h	Medium
35	clinic	Shree Ramachandra ENT Centre	12.5290096, 78.2037359	30 km/h	Medium
36	hospital	KVS Hospital	12.5277857, 78.2055049	30 km/h	Medium
37	hospital	Ga Heart Hospital	12.5287708, 78.2055336	30 km/h	Medium
38	clinic	Psychiatric Centre	12.529476, 78.205064	30 km/h	Medium
43	hospital	Shoolagiri Government Hospital	12.6669051, 78.015584	30 km/h	Medium

Crowded Spots

	type	name	coordinates	speed_limit	risk_level
0	school	KRP Matric. Hr. Sec School	11.4546193, 77.8142445	30 km/h	Medium
10	school	Fathima Hr Sec School	11.7410508, 78.0479272	30 km/h	Medium
21	college	Dhirajlal Gandhi College of Technology	11.7968014, 78.0562095	30 km/h	Medium
24	college	Laxminarayana Polytechnic College	12.0421277, 78.103203	30 km/h	Medium
26	school	Government Boys Higher secondary school	12.3047059, 78.2065495	30 km/h	Medium
27	school	Nalanda International Public School	12.4994979, 78.2235094	30 km/h	Medium
28	school	Don Bosco School Ground	12.505954, 78.2222003	30 km/h	Medium
39	college	Government Medical College	12.582332, 78.1496363	30 km/h	Medium
40	school	Crescent Matriculation Higher Seconddary School	12.6003724, 78.132787	30 km/h	Medium
41	college	Jeeva College of Nursing	12.6067058, 78.1191944	30 km/h	Medium
42	school	New Little Flower Matriculation Higher Secondary School	12.6607923, 78.0169012	30 km/h	Medium

Route Photos of Risky Spots



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr
Coordinates: 11.44077, 77.87533



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.44898, 77.87410



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.45352, 77.85698



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.46303, 77.84945



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.45545, 77.81812



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.45584, 77.81695



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.45656, 77.81709



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.49448, 77.88490



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.56352, 77.90078



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.59034, 77.90578



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.59328, 77.90809



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.63867, 77.94841



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.64217, 77.95200



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.64712, 77.95118



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.67858, 77.96834



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.69776, 77.97983



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.70179, 77.97900



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.71394, 77.99572

Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.73766, 78.03445



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.73651, 78.03868



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.74120, 78.04588



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.86805, 78.08353



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.88084, 78.08108



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.91159, 78.05857



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.93307, 78.06426



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.93545, 78.06130



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.93694, 78.05425



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.94867, 78.04680



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.95648, 78.04768



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.95803, 78.04964



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.96068, 78.04960



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.96370, 78.05142



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.96650, 78.04927



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 12.14025, 78.12819



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 12.14099, 78.12862



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 12.14056, 78.13957



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 12.17102, 78.16860



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 12.23131, 78.19497



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 12.24611, 78.19519



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 12.25200, 78.20034



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 12.31895, 78.20727



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 12.40026, 78.21775



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 12.40734, 78.22344



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 12.50342, 78.22352



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 12.54446, 78.20114



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 12.61917, 78.08303



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 12.63696, 78.07061



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 12.66145, 78.01543



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 12.66919, 78.01683



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 12.67465, 78.01432



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 12.67585, 78.01191



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 12.70105, 78.00458



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 12.70419, 78.00083



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 12.71008, 78.00038



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 12.71082, 77.99959



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 12.71395, 77.99898



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 12.71624, 77.99629



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 12.71751, 77.99579



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 12.75453, 77.97518



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 12.76098, 77.94755



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 12.75902, 77.94528



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 12.75966, 77.93945



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 12.75787, 77.93620



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 12.76227, 77.92663



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 12.76782, 77.92477



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 12.76728, 77.92147



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 12.76748, 77.91705
